

YI-LEI LI

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Summary

Software engineer with an **M.S. in Computer Science** (University of Delaware, May 2026) who ships tested, production-style systems: a deployed **React + TypeScript** app on **Node.js / PostgreSQL** with its own offline sync engine, a **Java 21 / Spring Boot 3** banking service with atomic transfers, and a **TypeScript** pipeline pulling 22,000+ records per run. Prior backend internship at Cathay Financial Holdings.

Technical Skills

Languages: Java, TypeScript/JavaScript (Node.js), Python, SQL, HTML5/CSS3
Frontend: React 19, React Router, Vite, Tailwind CSS, PWA (offline, service worker), IndexedDB (Dexie)
Backend: Spring Boot 3, Node.js (Hono), JPA/Hibernate, Drizzle ORM, REST APIs, pessimistic locking, Actuator
Databases: PostgreSQL (Neon, Row-Level Security), H2, DuckDB; window functions, CTEs
Testing & DevOps: JUnit 5, MockMvc, Vitest, Playwright, ESLint, Maven, CI/CD (GitHub Actions, Render), Git/GitHub
ML & AI: LLM integration (Anthropic Claude API), scikit-learn, XGBoost, SHAP, AI-assisted development

Projects

LeetCode Coach — Offline-First Interview-Prep Web App — React, TypeScript, Hono, PostgreSQL [\[Live\]](#) [\[GitHub\]](#)

- Full-stack **PWA**: a **React 19 + TypeScript** front end on IndexedDB over a **Node.js (Hono)** API on **PostgreSQL** (Drizzle ORM), with zod schemas shared by client and server; deployed on Render + Neon, and each deploy waits for CI to pass.
- Built an **offline-first sync engine**: every local write records an outbox entry in the same transaction; the server applies **last-write-wins** upserts with tombstones and assigns versions from a Postgres sequence under a **per-user advisory lock**, so a sync cursor never skips a change.
- Session auth with **script**-hashed passwords, SHA-256-hashed session tokens, HttpOnly host-prefixed cookies, CSRF checks, and rate limits; **Claude API** feedback on spoken explanations with JSON-schema output and a per-user daily quota. **140 Vitest** unit/API/sync tests and **17 Playwright** end-to-end tests on desktop Chrome and iPhone WebKit.

Banking Transactions API — Java 21, Spring Boot 3, PostgreSQL [\[GitHub\]](#)

- RESTful core-banking service with **atomic transfers** — both legs commit or neither does (@Transactional rollback); **ordered pessimistic row locks** (SELECT ... FOR UPDATE, ascending-id order) prevent double-spend and deadlock, and every movement writes an **immutable audit-ledger** row.
- 19 JUnit/MockMvc tests** (service rules + HTTP error contracts: 201/400/404/409) run in **GitHub Actions CI** on every push; **Actuator** health/metrics endpoints with a real DB connectivity check.

Job-Opening Monitor — Node.js, TypeScript, Supabase [\[GitHub\]](#)

- Scheduled **automation pipeline** pulling **22,000+ records per run** from 115 employer and job-board sources through **11 platform adapters** (Workday, Greenhouse, Lever, Ashby, iCIMS, Oracle Cloud, ...) behind one normalization layer, with bounded-concurrency fetches and per-source failure isolation.
- Idempotent** re-runs via a Postgres state table; runs unattended on a cron with Telegram + email alerts. Its Claude API screening stage **fails closed**: after exhausted API credit let 82 unscreened results out, errors now hold them back. **327 Vitest tests**.

quantlab — Multi-Market Backtesting & Research Engine — Python, pandas, NumPy [\[GitHub\]](#)

- Vectorized backtester for equities, crypto, and prediction markets with **next-bar execution** (no look-ahead bias), fee/slippage modeling, and **anchored walk-forward validation** reporting in-sample vs out-of-sample Sharpe per fold — a 1-hour breakout with **4.36** in-sample Sharpe scored **-0.89** out-of-sample and was rejected.

Education

University of Delaware

Master of Science in Computer Science — GPA 3.66/4.0

Newark, DE

Aug 2024 – May 2026

National Kaohsiung University of Science and Technology

Bachelor of Science in Electrical Engineering

Kaohsiung, Taiwan

2020

Experience

Independent Engineering Projects — self-directed, Newark, DE

May 2026 – Present

- Building and operating the systems above since completing the M.S. — most recently **LeetCode Coach** (September 2026) and eleven source adapters for the job-opening pipeline.

Cathay Financial Holdings Co., Ltd.

Jul 2023 – Nov 2023

Back-End Software Engineering Intern

Taipei, Taiwan

- Delivered a CRUD service (**Java + SQL**) on an **Agile** team and traced the firm's transaction-processing flow end to end — balance checks before transfers and per-account concurrency control (**ACID** integrity and isolation).

Graduate Research Assistant, Chang Gung University — **NeRF** 3D scene reconstruction

2021 – 2022

Undergraduate Research Assistant, NKUST IoT Lab — live sensor dashboard (**REST/WebSocket**)

2018 – 2020